

Assignment 4: Usability Testing & Analysis

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Submission Date: September 14, 2026

Project Title: RouteEase Mobile App

Design Tool: Figma Ecosystem

1. Usability Objectives & Testing Scope

Following the interactive layout phase of the **RouteEase** Smart Local Commute Assistant application, a professional usability diagnostic sequence was conducted. The main goal was to gather real feedback, evaluate user flows, and find interaction problems within the multi-modal transit journey.

2. Test Scenarios & Recruitment Parameters

A sample group of 5 target testers matching the user demographics was selected to execute two core tasks:

- Scenario A (Multi-Modal Search Loop):** Find a route from the home dashboard to an unfamiliar cross-city point using dynamic bus and rail components.
- Scenario B (Unified Checkout Node):** Move through the journey breakdown view and purchase a single combination transfer ticket (\$4.50) without errors.

3. Quantitative Performance Log

User interactions were observed in detail to track critical performance parameters, including task success rate, average time-on-task, and error frequency:

Evaluated Core Task Journey	Success Rate	Avg Time	Avg Errors	Friction Notes
Scenario A: Multi-Modal Search Flow	80%	42 seconds	1.4	Users missed recent route chevrons due to narrow layout boundaries.
Scenario B: Unified Checkout Node	100%	28 seconds	0.2	High task success; button contrast provided a strong call-to-action focus.

4. Actionable Design Improvement Matrix

Based on observation analysis, specific design improvements were implemented to eliminate interaction issues:

Identified Usability Gap	Testing Feedback & Observation Log	Implemented Layout Refinement
1. Interaction Boundary Friction	Users faced slight touch friction when clicking dashboard history items.	Expanded click target boundaries to a clean 48x48dp area matching accessibility standards.
2. High Cognitive Load	Active switching between routes created visual noise and crowded layouts.	Removed non-essential card visual elements to highlight transfer times and price structures.

5. Iterative Design & Pedagogical Takeaway

This usability testing process demonstrated that travel certainty is more critical than finding the absolute lowest price. Iterative testing models confirm that simple monochrome layouts help stakeholders evaluate content hierarchies and core navigation flows accurately before moving into high-fidelity production phases.